

State Minimization

Definition: $\equiv, \equiv_n$

Let $M = (K, \Sigma, \delta, s, F)$ be a DFA. Define $A_M \subseteq K \times \Sigma^*$ by $(q, w) \in A_M$ if and only if $(q, w) \vdash_M^* (f, e)$ for some $f \in F$. Two states $p, q \in K$ are called *equivalent*, written $p \equiv q$, if $(p, z) \in A_M \iff (q, z) \in A_M$ for every $z \in \Sigma^*$.
For $n \geq 0$, we say that $p \equiv_n q$ if the same holds for every $z \in \Sigma^*$ with $|z| \leq n$.

Q1. Let M be a DFA, and let p and q be two of its states.

- (a) What does $p \equiv_0 q$ mean?
- (b) Give an example of such an M for which $p \equiv_0 q$ but $p \not\equiv_1 q$. Prove your answer.

Lemma

For any two states $p, q \in K$ and any integer $n \geq 1$, $p \equiv_n q$ if and only if (a) $p \equiv_{n-1} q$, and (b) for every $a \in \Sigma$, $\delta(p, a) \equiv_{n-1} \delta(q, a)$.

Algorithm Computing the equivalence classes of $\equiv$

Require: DFA $M = (K, \Sigma, \delta, s, F)$

- 1: Initially, the classes of $\equiv_0$ are F and $K - F$
- 2: Let $n = 0$
- 3: **repeat**
- 4: $n = n + 1$
- 5: Compute the classes of $\equiv_n$ from those of $\equiv_{n-1}$
- 6: **until** $\equiv_n$ is the same as $\equiv_{n-1}$
- 7: **return** the classes of $\equiv_n$

Q2. Let M be the DFA over $\{a, b\}$ given by the transition table below. Its start state is q_0 , and its only final state is q_6 . We will construct a DFA with the fewest possible states accepting $L(M)$.

q	q_0	q_1	q_2	q_3	q_4	q_5	q_6	q_7
$\delta(q, a)$	q_1	q_2	q_6	q_4	q_5	q_6	q_6	q_7
$\delta(q, b)$	q_3	q_3	q_3	q_3	q_3	q_3	q_6	q_7

- (a) Identify and remove the states unreachable from q_0 .
- (b) A student claims that $q_0 \equiv q_1$ because neither state is final and an a -transition from either leads to a nonfinal state. What is wrong with this argument?
- (c) Compute the equivalence classes of $\equiv_0$ and $\equiv_1$. Then use the algorithm above to compute the classes of $\equiv$. Explain why $\equiv_n$ is the same as $\equiv$ when the algorithm stops.
- (d) Construct a DFA M' with the equivalence classes of $\equiv$ as states, start state $[q_0]$, final states $\{[q] : q \in F\}$, and transition function $\delta'([q], a) = [\delta(q, a)]$. Give its transition table.
- (e) Describe $L(M)$ in plain English.

Q3. Let M' be the DFA constructed in Q2.

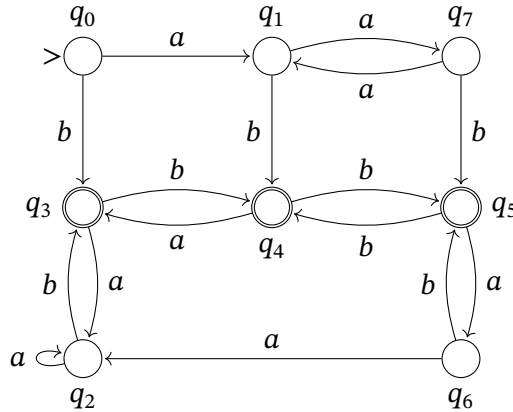
- Show that the transition function δ' is well defined.
- Prove that $L(M') = L(M)$.

Q4. Let $M := (\{q_0, \dots, q_8\}, \{a, b\}, \delta, q_0, \{q_2, q_5, q_8\})$, where δ is given below.

q	q_0	q_1	q_2	q_3	q_4	q_5	q_6	q_7	q_8
$\delta(q, a)$	q_1	q_7	q_1	q_2	q_3	q_6	q_8	q_1	q_7
$\delta(q, b)$	q_0	q_4	q_4	q_5	q_0	q_0	q_5	q_4	q_4

- Use the algorithm above to compute the equivalence classes of $\equiv$ for M . Show the classes obtained in each iteration.
- Draw a minimal DFA for $L(M)$. Describe $L(M)$ in plain English.

Q5. Let M be the DFA over $\{a, b\}$ shown below.



- Construct a minimal DFA M' accepting $L(M)$.
- Delete the b -transitions from M' . Is the resulting DFA over $\{a\}$ minimal? Explain.
- Let s and δ be the start state and transition function of M' . Change the start state of M' to $\delta(s, b)$. Is the resulting DFA minimal? Explain.

Definition: $\sim_M$

Let $M = (K, \Sigma, \delta, s, F)$ be a DFA. Two strings $x, y \in \Sigma^*$ are *equivalent with respect to M* , written $x \sim_M y$, if there is a state $q \in K$ such that $(s, x) \vdash_M^* (q, e)$ and $(s, y) \vdash_M^* (q, e)$.

Q6. Let R and S be equivalence relations on a set X , with $R \subseteq S$ (R is a *refinement* of S). The *quotient* of S by R , denoted S/R , is a relation on the equivalence classes of R , defined by

$$([x]_R, [y]_R) \in S/R \quad \text{if and only if} \quad (x, y) \in S.$$

- What does $R \subseteq S$ mean for the equivalence classes of R and S ?
- Let X be the students at BITS, R the same-hostel relation, and S the same-campus relation. Describe the equivalence classes of R and the relation S/R in plain English.
- Let $X = \{1, 2, 3, 4, 5, 6, 7\}$. The classes of R are $\{1, 2\}$, $\{3\}$, $\{4, 5\}$, and $\{6, 7\}$; the classes of S are $\{1, 2, 3\}$ and $\{4, 5, 6, 7\}$. Which classes of R are related by S/R ?
- Show that S/R does not depend on which representatives are chosen from the classes of R .
- Show that S/R is an equivalence relation.

- (f) $\sim_M$ is a refinement of $\approx_{L(M)}$. Describe $\approx_{L(M)} / \sim_M$ in plain English.
- (g) For the DFAs M and M' from Q2, find strings x, y such that $x \sim_M y$ but $x \not\approx_{L(M)} y$. Compare $\sim_{M'}$ with $\approx_{L(M)}$.

Q7. Let M be a minimal DFA. Let M' be the DFA obtained on interchanging the final and nonfinal states of M . Is M' also minimal? Prove your answer.

Q8. Let L be a language over an alphabet Σ . Show that all minimal DFAs over Σ that accept L are isomorphic (the same up to a renaming of states).

Q9. Let M_1 and M_2 be DFAs over the same alphabet. How could you use state minimization to decide whether $L(M_1) = L(M_2)$? Is comparing the resulting numbers of states sufficient? Give an example.